

DRISHYAM

From Footage to Action

Database Schema Document

PostgreSQL Schema for Multi-tenant Real-time System

Document ID: DRISHYAM-DB-001

Version: 1.0 | 13 September 2026

Confidential – For Development Use Only

1. Design Principles

- Multi-tenant isolation via `organization_id` on all tenant data
- Soft deletes where auditability is required
- JSONB for flexible config and metadata
- Timestamps (`created_at`, `updated_at`) on all major tables
- UUID primary keys for distributed-friendly IDs

2. Core Tables

organizations

- `id` UUID PK
- `name` VARCHAR(255) NOT NULL
- `slug` VARCHAR(100) UNIQUE
- `settings` JSONB DEFAULT '{}'
- `retention_days` INT DEFAULT 30
- `created_at`, `updated_at` TIMESTAMPTZ

users

- `id` UUID PK
- `email` VARCHAR(255) UNIQUE NOT NULL
- `password_hash` VARCHAR(255)
- `full_name` VARCHAR(255)
- `is_active` BOOLEAN DEFAULT true
- `last_login_at` TIMESTAMPTZ
- `created_at`, `updated_at` TIMESTAMPTZ

organization_members

- `id` UUID PK
- `organization_id` UUID FK → organizations
- `user_id` UUID FK → users
- `role` VARCHAR(50) NOT NULL -- OrgAdmin, SiteManager, Operator, Viewer
- `site_ids` UUID[] -- optional scope; null = all sites
- `created_at` TIMESTAMPTZ
- UNIQUE(`organization_id`, `user_id`)

sites

- `id` UUID PK
- `organization_id` UUID FK → organizations
- `name` VARCHAR(255) NOT NULL
- `address` TEXT
- `timezone` VARCHAR(50) DEFAULT 'Asia/Kolkata'
- `settings` JSONB DEFAULT '{}'
- `created_at`, `updated_at` TIMESTAMPTZ

cameras

- `id` UUID PK
- `site_id` UUID FK → sites
- `organization_id` UUID FK → organizations
- `name` VARCHAR(255) NOT NULL
- `rtsp_url` TEXT NOT NULL
- `onvif_host` VARCHAR(255)
- `status` VARCHAR(30) DEFAULT 'unknown' -- online, offline, degraded
- `last_seen_at` TIMESTAMPTZ

- config JSONB DEFAULT '{} ' -- zones, thresholds, enabled_classes
- edge_node_id UUID -- optional
- created_at, updated_at TIMESTAMPTZ

alerts

- id UUID PK
- organization_id UUID FK
- site_id UUID FK
- camera_id UUID FK
- alert_type VARCHAR(50) NOT NULL -- weapon, violence, intrusion, loitering
- severity VARCHAR(20) NOT NULL -- critical, high, medium, low
- confidence FLOAT
- status VARCHAR(30) DEFAULT 'new' -- new, acknowledged, resolved, false_positive
- detected_at TIMESTAMPTZ NOT NULL
- clip_path TEXT
- thumbnail_path TEXT
- metadata JSONB DEFAULT '{} '
- acknowledged_by UUID FK → users
- acknowledged_at TIMESTAMPTZ
- resolved_at TIMESTAMPTZ
- created_at TIMESTAMPTZ

alert_actions

- id UUID PK
- alert_id UUID FK → alerts
- user_id UUID FK → users
- action VARCHAR(50) -- acknowledge, resolve, false_positive, escalate, comment
- note TEXT
- created_at TIMESTAMPTZ

audit_logs

- id UUID PK
- organization_id UUID
- user_id UUID
- action VARCHAR(100) NOT NULL
- resource_type VARCHAR(50)
- resource_id UUID
- details JSONB
- ip_address INET
- created_at TIMESTAMPTZ

edge_nodes

- id UUID PK
- organization_id UUID FK
- name VARCHAR(255)
- last_heartbeat_at TIMESTAMPTZ
- status VARCHAR(30)
- config JSONB
- created_at, updated_at TIMESTAMPTZ

3. Key Indexes

- alerts (organization_id, detected_at DESC)
- alerts (site_id, status, detected_at DESC)
- alerts (camera_id, detected_at DESC)

- cameras (site_id)
- organization_members (user_id)
- audit_logs (organization_id, created_at DESC)

4. Relationships Summary

Organization 1—N Sites 1—N Cameras. Organization N—N Users via organization_members. Alerts belong to Org + Site + Camera. Alert actions and audit logs reference users and resources.